

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 4.2.A** — Identify key characteristics of a parametric planar motion function related to position: find the horizontal and vertical extrema from the max/min values of $x(t)$ and $y(t)$, and locate x - and y -intercepts of the curve from the zeros of $y(t)$ and $x(t)$.

Vocabulary & Key Concepts

horizontal extremum: The **maximum** or **minimum** value of $x(t)$; gives the rightmost or leftmost position of the particle.

vertical extremum: The maximum or minimum value of $y(t)$; gives the highest or lowest position of the particle.

y -intercept (of parametric curve): A point on the curve where $x(t) = 0$; found by solving $x(t) = 0$ for t , then evaluating $y(t)$ at those values.

x -intercept (of parametric curve): A point on the curve where $y(t) = 0$; found by solving $y(t) = 0$ for t , then evaluating $x(t)$.

Hook

Launch: A ball's position (feet) at time t seconds is $x(t) = -16t^2 + 48t$ and $y(t) = 12t - t^2$.

Q1. Which component do you examine to find how high the ball goes? What equation do you solve to find when the ball lands?

Examine $y(t)$ for height. To find when the ball lands, solve $y(t) = 12t - t^2 = 0$.

LO 4.2.A — Planar Motion: Horizontal and Vertical Extrema

Example: Let $f(t) = (x(t), y(t)) = (t^2 - 4t + 3, -t^2 + 2t)$ for $0 \leq t \leq 4$.

Step 1 — Build the table.

t	$x(t) = t^2 - 4t + 3$	$y(t) = -t^2 + 2t$	(x, y)
0	3	0	(3, 0)
1	0	1	(0, 1)
2	-1	0	(-1, 0)
3	0	-3	(0, -3)
4	3	-8	(3, -8)

Step 2 — Find horizontal extrema of $x(t) = t^2 - 4t + 3$.

$x(t)$ is a **upward-opening** parabola. Vertex at $t = -\frac{b}{2a} = -\frac{-4}{2} = 2$

Minimum value of x : $x(2) = -1$ (particle's **left**-most position)

Maximum value of x on $[0, 4]$: check endpoints. $x(0) = 3$, $x(4) = 3$. Maximum is 3 (particle's **right**-most position).

Step 3 — Find vertical extrema of $y(t) = -t^2 + 2t$.

Vertex at $t = -\frac{b}{2a} = 1$.

Maximum value of y : $y(1) = 1$ (particle's **high**-est position)

Minimum of y on $[0, 4]$: check endpoints. $y(0) = 0$, $y(4) = -8$. Minimum is -8 .

LO 4.2.A — Axis Intercepts of a Parametric Curve

Key rule: Zeros of $x(t)$ give **y**-intercepts of the curve. Zeros of $y(t)$ give **x**-intercepts.

Example (continued): $f(t) = (t^2 - 4t + 3, -t^2 + 2t)$, $0 \leq t \leq 4$.

Find the y-intercepts (where $x(t) = 0$):

Set $t^2 - 4t + 3 = 0$. Factor: $(t - 1)(t - 3) = 0$.

$t = 1$ or $t = 3$.

Evaluate y at each: $y(1) = 1$, $y(3) = -3$.

y -intercepts of the curve: $(0, 1)$ and $(0, -3)$.

Find the x-intercepts (where $y(t) = 0$):

Set $-t^2 + 2t = 0$. Factor: $-t(t - 2) = 0$.

$t = 0$ or $t = 2$.

Evaluate x at each: $x(0) = 3$, $x(2) = -1$.

x -intercepts of the curve: $(3, 0)$ and $(-1, 0)$.